

World's first holographic flexible smartphone

Scientists claim to have developed the world's first holographic flexible smartphone that lets users interact with 3D videos and images without any headgear or glasses. They have named the device HoloFlex. It is capable of rendering 3D images with motion parallax and stereoscopy to

multiple simultaneous users without head tracking or glasses.

HoloFlex features a 1920×1080 full-high-definition flexible organic light emitting diode (FOLED) touchscreen display. Images are rendered into 12-pixel-wide circular

blocks showing the full view of the 3D object from a particular viewpoint. These pixel blocks project through a 3D-printed flexible micro-lens array consisting of more than 16,000 fisheye lenses. The resulting 160×104 resolution image allows users to inspect a 3D object from any angle simply by rotating the phone.

Among other applications of HoloFlex is holographic gaming. For example, in a game like Angry Birds, users can bend the side of the display to pull the elastic rubber band that propels the bird. When the bird flies across the screen, the holographic display makes the bird literally pop out of the screen in 3D.



HoloFlex offers a completely new way of interacting with a smartphone (Image courtesy: <http://naked-science.ru>)

Mini drone uses electrode patch to stick to surfaces

A robotics team at Harvard University, USA, has developed a method that would allow their insect-size flying robot, dubbed RoboBee, to conserve energy midflight, much as bees, bats and birds do. By attaching a shock-absorbing mount and a patch that conducts electricity, the researchers were able to direct the tiny robot to perch on a variety of surfaces and then take off again. When activated, the electrical charge held RoboBee in place, much like how a balloon would stick to a wall after it was rubbed against a wool sweater. Terminating the charge enabled the robot to detach from the surface and fly away.

The robot made its debut in 2013. It was the first robotic insect that was capable of hovering, and it was modified for the new study to allow it to land midflight.



RoboBee weighs about 100 milligrams and is 20 millimetres tall—about the size and weight of an actual bee (Image courtesy: www.dailymail.co.uk)